

- 1 The times taken for students at a college to run 200 m have a normal distribution with mean μ s. The times, x s, are recorded for a random sample of 10 students from the college. The results are summarised as follows, where $\bar{x}$ is the sample mean.

$$\bar{x} = 25.6 \qquad \sum (x - \bar{x})^2 = 78.5$$

- (a) Find a 90% confidence interval for μ . [4]

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A test of the null hypothesis $\mu = k$ is carried out on this sample, using a 10% significance level. The test does not support the alternative hypothesis $\mu < k$.

- (b) Find the greatest possible value of k . [3]

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Use a t -test, at the 5% significance level, to test whether Manet's claim is justified, stating any assumption that you make. [8]

This image shows a full page of a worksheet designed for handwriting practice. It features approximately 20 horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.